Banya Framework Comprehensive Report

Official Name: Banya Framework

Alias: Buddha's Palm Framework

Classification: Axiom-Based Science Mining Engine

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> Version: v1.9 (2026-10-13)

Version change history (v1.5~v1.9)

> v1.9 changes: definition of the axioms settled -- the 15 axioms were completed by design first, and mining shapes the axiom system like weights by adding propositions while the axioms stay fixed. recovering a known value in its original domain is post-hoc agreement, a new relation obtained in an empty domain is a discovery, and the verdict comes from empty-domain discoveries and predictions of unmeasured values (How the axioms were made). The fitting-impossibility block reduced to "Fitting versus Learning"; shake experiments removed. All Common Misconceptions moved into the main page (old address forwards automatically) with the Usage Cautions absorbed into them, Science Mining Manual restructured, derivation demos extended to the four Axiomatic Model papers. one proposition added to Axiom 4 (Level-Spacing Class). The number system is fixed by the orthogonality of the Banya equation and the level-spacing class by cost -- irreversible operations take the complex class, reversible operations the real class, and the Cl(3) bivector rotation of juida the quaternion class. Added a date column to the hypothesis library and prediction tables, and a creation date at the top of each card. Dates follow the first registration date in the documents; where none exists, the card creation date.

> v1.8 changes: Clifford form proposition reorganized -- distinguished into Cl(7) structure (data type 128 = 2^7) and Cl(3) operator (Axiom 2 Proposition "Juida (Writing) Is a Cl(3) Bivector Rotation"). Cl(4) registered only as structural notation -- simultaneous firing on the bit-transition diagram (Axiom 15) collides with CAS sequential enforcement (Axiom 5), so the absence of a Cl(4) operator is made explicit. Added the "Complementarity of the Banya Equation and the Bit-Transition Diagram" proposition to Axiom 2 -- CAS cannot be seen in the Banya Equation's geometric representation; its coordinates appear only in the bit-transition diagram. Workbench proposition corrected -- CAS resides in the norm space as an opcode operator. Physics-borrowed vocabulary (pressure, spacetime, measurement) removed; δ-as-migrator expressions purged; "firing" restricted to the δ firing bit (Axiom 15) only. Bulk auto-linking of axiom references (471 in-body "Axiom N" mentions activated). Unified h3/h4 font sizes (1.75em / 1.5em). AI mining box now prioritizes live pages (ubmscoin.github.io/banya/en/banya.html) over PDF in the 3-step procedure. The "Axiom-Derived Formula Summary" and the "Axiom System Summary Tables" were removed from axiom.html for body cleanup.

> v1.7 changes: Added 1 proposition to Axiom 1 (7-Axis Orthogonality in Clifford Form Cl(7)). Exposes the algebraic structure of data type 128 = 2^7 as the Cl(7) multivector space. Quaternion proposition application target also expanded -- not only spatial 3 axes but all 3-axis orthogonal groups (including CAS 3 axes). Mining tools expanded: Cl(7) multivector representation of 7-bit cost/dimension/movement

> v1.6 changes: Added 2 propositions to Axiom 1 (Banya Equation in Unit Circle Form, Subframe Orthogonality in Unit Circle / Unit Sphere Form -- Quaternion). Added 1 proposition to Axiom 4 (Reading Cost in Norm). All are structural consequences without additional assumptions -- 0 new axioms. Mining tools expanded: phase (-1)^n, Euler form, quaternion rotation, explicit combination rules for cost reading vs norm reading

> v1.5 changes: Complete overhaul of the AI 3-step paste procedure -- significantly improved AI learning success rate. Strengthened the guidance structure so AI operates as a mining tool rather than falling into theoretical analysis

Each card below is clickable in this PDF — jump to the live page
https://ubmscoin.github.io/banya/en/banya.html#quicknav

AI-Driven Mining Guide
3-step paste procedure

15 Axioms
Banya equation + axiom system

Derivation Demo
Axiomatic Model papers

Science Mining Manual
5-step recursive substitution

Sub-Reports
28 discovery reports index

Common Misconceptions
The axioms were learned + 11 misconceptions

Hypothesis Library
Discoveries + hypotheses + predictions 1,700+

Unique Predictions
600+ staked before measurement

118 Compatibility Verification
Physics equations × Banya Framework

Author Template
Discovery/hypothesis report form

Inventor's Reflection
Design starting point

Intro Screen
Banya Framework Sandbox

What is the Banya Equation / What is the Banya Framework

The Banya Framework is an axiom-based science mining engine. It starts from 15 axioms, derives physical constants and laws in the forward direction, and judges the results against measured values and falsification conditions.

Item	Content
Input	15 axioms, measured physical constants, library cards already judged
Process	Five-step recursive substitution (Banya equation, norm substitution, constant substitution, domain conversion, finding)
Output	One card. The verdict is one of discovery, hypothesis, prediction, retired (hypothesis library, unique predictions)
Criteria	Error against measured values, and falsification conditions attached to values not yet measured
What it is not	Not a Theory of Everything and not a replacement for existing physics. The one-line Banya equation by itself produces no numbers

The **Banya Equation** and the **Banya Framework** are different.

Banya Equation, a 4-axis orthogonal norm in one line

The Banya Equation comes from the Heart Sutra (Banya心經). "Everything changes except change itself" becomes δ (total variation), and "form is emptiness, emptiness is form" (色卽是空 空卽是色) is rewritten as the orthogonality of classical (time+space) and quantum (observer+superposition). The insight is transcribed directly into the language and formulas of computer science:

$$\delta^2 = (time + space)^2 + (observer + superposition)^2$$

δ: change | time: time | space: space | observer: observation | superposition: superposition

The Banya equation is one line made of 4 words and 2 squares. It is a structural declaration that all change in the universe is the norm of 4 axes. Every derivation starts here. The Banya equation declares only topology (what is orthogonal to what); distance, sign and unit are fixed when constants are substituted (Axiom 1 Proposition).

Banya Framework, the engine that runs the equation

It is a tool system that takes the Banya Equation as its engine, fixes **CAS (Read-Compare-Swap)** as the sole atomic operator, and bounds resources by finite cost (≤+1, Axiom 4). It is closed by 15 axioms and propositions and imports 0 external equations.

Why CAS is the sole operator. The δ in the Banya Equation deals only with one change. Change is defined only when the pre-change state and the post-change state are compared at the same time. CAS is the only atomic operator that handles two states inside one operation, and therefore the only operator that brings forth order and change. Regardless of which domain (time, space, observer, superposition) CAS operates through, the atomic operator at the bottom is the same CAS.

Framework = Banya Equation + 1 CAS + description DOF 9 (fixed) + 1 kind of 1D ring (fixed) + cost

The equation declares the structure, and the framework substitutes constants into that structure to produce numbers.

The whole picture at a glance

Item	Status
Discoveries (D), hypotheses (H), predictions (P)	More than 1,700 accumulated in the Hypothesis Library. All derived from the same axioms through forward chains
Unique predictions	More than 600 in Predictions. Predictions of unmeasured values with falsification conditions, registered before measurement; graded after measurement
Compatibility with existing physics	118 equations checked, FAIL 0. Includes $E=mc^2$, uncertainty principle, black hole entropy
Forward derivation of Standard Model constants	22 constants

Derivation demos

Axiomatic Model papers deriving different physical domains on the same axiom set. Each paper marks post-hoc reproductions apart from falsifiable predictions.

Part	Subject	Key result	Full text
I	Fine-structure constant α	The axioms' cost classification forces the signature (5,2). $1/\alpha = 137.036082$ (error 6×10^{-7}), $\sin^2\theta_W = 0.23122$ (4×10^{-6}), $\eta_B = 6.14\times10^{-10}$ (0.5σ from Planck)	PDF (49 pp.) DOI
II	Wheeler-DeWitt equation	The four ingredients of the equation are forced by the bit reading (0,1,0,1) of the 4 domain axes. The $6 = T(3) = 7-1$ independent components and the (5,1) signature of the DeWitt metric are derived by two independent paths	PDF (26 pp.) DOI
III	Black hole thermodynamics	The 4 in $S = A/4$ is the domain count 4 and the 4 in the QNM damping spacing 1/4 is the ball value 4: different quantities (distinguishable in higher dimensions). It obtains the evaporation coefficient $5120 = 10\times2^9$ and reproduces $r_s = 2GM/c^2$. 4 unique predictions with falsification conditions	PDF (18 pp.) DOI
IV	Koide formula	Identifies the amplitude $r^2 = 2$ and the phase angle 2/9 with registered numbers of the axioms (post-hoc reconstruction). Shows as a consequence of the axioms the observed fact that the Koide relation holds only at pole masses and breaks at higher scales. Deviation-size candidate $\alpha(\mu)/(2\pi)$	PDF (24 pp.) DOI

Reports built the same way accumulate in the Hypothesis Library, and values not yet measured are separated, with falsification conditions, into Unique Predictions, awaiting experimental verification.

Next path: read 15 Axioms, the derivation demos, the Hypothesis Library (more than 1,700 discoveries, hypotheses and predictions), and the Science Mining Manual, in that order.

Fitting versus Learning

How the axioms were made

The axioms were learned. The 15 axioms were completed by design before mining. While more than 1,700 physics equations were then mined, the axiom system was shaped like the weights of a trained model. The 9 rounds of shaping up to v1.9 were done by the hand of the axiom author, keeping the 15 axioms fixed and adding propositions.

Agreement with already known values, recovered deductively in the domain where they already lived, is therefore a necessary result of the learning, and the framework regards it as post-hoc agreement. By contrast, a relation or value that did not exist before, obtained by converting into an empty domain, is judged a discovery. That 15 fixed axioms cover that many equations is the achievement of the learning; the discriminating power lies in empty-domain discoveries and in predictions of unmeasured values made with the learned axioms (unique predictions).

Beyond wiring up what has been seen and deducing from it, the work of the axioms is to infer what was never seen. The learned axioms have staked more than 600 values not yet measured, each with falsification conditions. Every card carries its creation date and the public repository keeps its history, so the record proves these inferences were made before measurement. That 15 fixed axioms extend inference this far into unseen territory is where the value of the Banya Framework shows.

Grading, however, can only happen after measurement. Having inferred what was not seen does not mean every inference is right. Each prediction is judged by experiment one at a time, and wrong ones are kept with the reason for retirement attached. Some predictions have already been retired by the framework itself, through falsification or internal inconsistency.

It is announced in advance that AI will hereafter shape the weights of the axioms by itself, on the basis of mining results. At that stage the axioms are to be wrapped in the Lean theorem prover.

This learning differs from free-parameter fitting in 2 ways.

1. **The weights are integers.** There are no adjustable continuous knobs. The weights are the integer lists registered in Axiom 9 (structural 1, 2, 3, 4, 7, 9, 16, 30, 128, 137 / cost 1, 2, 3, 4, 5, 9, 13).
2. **The weights are shared.** The same integers recur in cards from different domains: masses, mixing angles, cosmology, black holes. They are not tuned card by card, so changing one integer throws cards in several domains off together.

At a glance

#	Misconception	Fact
1	It is a Theory of Everything	It is a framework, not a theory. It declares structure and locates existing theories
2	It replaces existing physics	Existing equations work unchanged inside the framework
3	4 axes = 4D spacetime	Spacetime is the time, space side. observer and superposition are not spacetime
4	δ is energy	Energy is only one way of measuring δ
5	+ is addition	+ declares orthogonality
6	An observer axis is unscientific	observer is the structural element that receives the Compare branch
7	The units are wrong	Axes are names. Units attach when constants are substituted
8	Minkowski uses −, why +	The equation is topology, not metric. Metric signs come after substitution, and the signature (5,2) is forced by the axioms
9	It is circular	It starts from the axioms and arrives at existing equations. Inserting back-calculated measurements is void
10	No new predictions	There is a list of unique predictions with falsification conditions
11	It is relabeling	Changing domain yields values renaming cannot produce

Common Misconceptions

Misconception 1. The Banya equation is a Theory of Everything (TOE)

Fact: the Banya equation is not a theory but a framework.

A theory explains "why" and predicts specific numbers. A framework defines "how far" and checks whether existing theories fit inside it.

> Theory: $E = mc^2$ ==> 1 kg of mass gives 9×10^{16} J of energy (numerical prediction)
> Framework: $E = mc^2$ lives in the classical bracket (time, space) (location check)

Example: the Pythagorean theorem declares a structure, "the square of the hypotenuse equals the sum of the squares of the other two sides." It does not say whether a side is 3 or 5. The Banya equation is the same.

The one-line Banya equation by itself produces no numbers. Numbers are obtained by substituting constants into the framework, so one says "derived with the Banya Framework," not "predicted by the Banya equation." Saying the equation predicts confuses it with a theory.

Misconception 2. The Banya equation replaces existing physics

Fact: existing physics equations work unchanged inside the Banya Framework.

Einstein's $E^2 = (mc^2)^2 + (pc)^2$ is an app in the classical bracket. The Schrödinger equation is an app in the quantum bracket. The framework does not replace the apps; it describes the OS they run on.

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Before: relativity app and quantum app separate, incompatible
Banya Framework: both apps run on the same OS. The two brackets are orthogonal, so no conflict
```

Example: Excel did not disappear when Windows arrived. Excel runs on Windows.

Misconception 3. Four axes means four-dimensional spacetime

Fact: spacetime is the time, space side of the classical bracket. The other two axes (observer, superposition) are the quantum bracket and are not spacetime.

```
Classical bracket: time, space ==> spacetime (Einstein's domain)
Quantum bracket: observer, superposition ==> quantum state (Heisenberg's domain)
The two are orthogonal ==> no need to merge
```

The observed 3+1 dimensions come from the single space axis unfolding into three components. Axiom 2 Proposition text: "The space axis is unfolded into 3 components (x, y, z) by the action of the CAS 3-axis orthogonal locks (R_LOCK, C_LOCK, S_LOCK)." And the Wyler appendix of the axiom document: "The subframe (3,1) is observable spacetime."

Example: a car's speedometer and fuel gauge are independent dials. The fuel gauge does not change because the speedometer rises. Yet both describe the state of the car.

Misconception 4. δ is energy

Fact: energy is only one way of measuring δ. Seen from inside the FSM, δ appears as change; in Axiom 15 the identity of δ is firing.

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δ = change (as seen inside the FSM, invariant)
Energy = one way of measuring change
Distance = another way of measuring change
Probability = another way of measuring change
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Axiom 15 Proposition text: "In Axioms 1-14, δ was "change" -- the left-hand side of the Banya equation, the 4-axis norm, the equality sign. In Axiom 15, δ's true identity is revealed: it is not change but firing. Change is the result produced by the FSM, and firing is the cause that wakes the FSM."

Example: whether you measure distance in km or miles, the distance is the same. Whether you measure δ as energy or as probability, δ is the same.

Misconception 5. The + sign in the equation is addition

Fact: + is a structural declaration that two things are orthogonal. It does not mean adding numbers.

Axiom 1 Proposition text: "Therefore the + sign must not be read as addition. The + is a topological declaration that the two brackets are orthogonal, not a metric instruction to sum squared distances."

$$(\text{time} + \text{space})^2 = \text{time}^2 + \text{space}^2$$

Orthogonal, so there is no cross term

Example: going 3 km east and 4 km north on a map gives a straight-line distance of 5 km. Not 3+4=7 but √(9+16)=5. Orthogonal quantities combine by Pythagoras.

Orthogonal means independent, not merged. Trade-offs happen only among axes in the same bracket (time² + space² = c²); axes from different brackets (time and observer) are never operated on directly in one equation.

Misconception 6. Putting observer as an axis is unscientific

Fact: observer is the structural element that receives the result of CAS's Compare. It is not a metaphor that brings in consciousness or people.

Axiom 1 text: "Without observer, there is nowhere to receive the result of CAS's Compare. Without superposition, there is nothing for CAS to reference. Both are structural necessities for the Banya Equation to operate, not figurative rhetoric borrowed from quantum mechanics."

Background: quantum mechanics has not solved the measurement problem for 100 years. The Copenhagen interpretation, many-worlds, and decoherence theory have all failed to answer "why does observation change the result." The Banya Framework moved observation from "something to be explained" to "an axis of the structure," the same strategy as Einstein defining gravity as the curvature of spacetime.

Einstein: don't know what gravity is ==> define it as spacetime curvature (success)
Banya Framework: don't know what observation is ==> define it as an independent axis (118 physics equations checked for compatibility)

Misconception 7. The units are wrong (it's neither SI nor natural units)

Fact: time, space, observer, superposition in the Banya equation are names of axes. They are not m (meters), s (seconds) or J (joules). Units attach when constants are substituted into the norm.

Axiom 1 Proposition text: "Metric (constant substitution) : sign and unit and distance = decided when constants are substituted"

Before substitution	After substitution	Unit
Classical bracket norm $\|\|C\|\|$	$\|\|C\|\| = c$	m/s
Quantum bracket norm $\|\|Q\|\|$	$\|\|Q\|\| = \hbar$	J*s
Relation of the two brackets	Orthogonal	Different units are never summed as numbers (Misconception 5)

Wrong: "time is seconds and space is meters, how can you add them"
Right: time and space have no units yet. Put c into the norm and both share the unit of c

Example: the word "distance" itself has no unit. It can be measured in km, miles or light-years. The axes of the Banya equation are likewise only names, and units attach the moment constants are inserted. Whether you use natural units (c = ħ = 1) or SI, the orthogonal structure does not change. Units are not a property of the framework but the user's choice.

Misconception 8. The signs are wrong (Minkowski uses − but you use +)

Fact: the Banya equation is topology, not metric. Minkowski's − is a sign of the metric, and the metric arises only after constants are substituted. These are different layers.

Axiom 1 Proposition text: "The distances between axes, the angles, and the signs are not yet fixed. That is, this equation is not a metric but a topology."

Layer	What it fixes	Example
Topology (Banya equation)	What is orthogonal to what	+ = orthogonal axes in the same bracket
Metric (constant substitution)	Sign, unit, distance	Expanding $\|\|C\|\|^2 = c^2$ gives $(ct)^2 - x^2 - y^2 - z^2 = ds^2$

Nor is the sign chosen at will. The signature of the metric is forced by the axioms. Among the 7 axes (4 domain axes + 3 CAS stages), assigning positive sign to the 5 irreversible cost-bearing axes and negative sign to the 2 reversible zero-cost axes (observer, superposition) gives the signature $(5, 2)$. Of the 8 possible partitions, only this one is compatible with the axioms. This is the $SO(5, 2)$ that Wyler chose arbitrarily in 1969, and its subframe $(3, 1)$ is the Minkowski spacetime we observe. The full derivation is in Science Mining Method, Sample 1 and the paper PDF.

Signs such as $i^2 = -1$ in the unit-circle, quaternion and Clifford forms of the axiom document are not metric either. Axiom 1 Proposition text: "Do not confuse a visible sign for a metric -- the sign of a metric is fixed only when constants are substituted."

Example: labeling a drawer "socks" and the way you fold socks are different matters. The Banya equation is the drawer label; the sign is how the socks are folded. The label does not need folding instructions.

Misconception 9. It is circular reasoning ($E_p = m_p c^2$ is a definition, not a derivation)

Fact: the direction is the opposite. It starts from CAS cost (Axiom 4) and arrives at existing physics equations. It does not take existing equations and put them into the framework.

```
Circular direction: know E = mc² ==> put it into the Banya Framework ==> "you already knew that"
Actual direction:  CAS cost (Axiom 4: cost = ħ , record = spacetime) ==> expand ==> E = mc² comes out
```

Running it with CAS cost set as a hypothesis gives the same result. A different starting point is not circular. Recovering an existing physical value in its original domain is post-hoc agreement; the hypothesis is judged by empty-domain discoveries and predictions of unmeasured values (see How the axioms were made). The mining rules also block circularity: inserting a back-calculated measured value and deriving the same value is void (Science Mining Method, Ch. 2).

Misconception 10. There are no new predictions (it merely reinterprets existing values)

Fact: there is a list of unique predictions giving values not yet measured, with falsification conditions. If wrong, they are retired.

Example: P-641 predicts that the neutrino is a Majorana particle and that the effective mass of neutrinoless double beta decay is $1.47 \sim 3.67$ meV. If a next-generation experiment sees a signal above 5 meV, it is retired.

Another route to something new is the empty domain. Each physics equation leaves some domains unused, and what would come out on converting into them can be written down in advance. When such a path actually closes, it becomes a discovery. Even if the quantity reached is already known, the relation that reaches it through an empty-domain path is new.

```
Coulomb's law: F = kq₁q₂/r² ==> uses only the space domain
Empty domains: observer, superposition, time
Converting: Larmor radiation, electromagnetic decoherence rate and entanglement energy are expected
```

The appendix (compatibility of 118 equations + empty-domain expectations) writes down each equation's empty-domain expectations in advance, and the paths that closed are registered as cards and marked as discoveries or hypotheses.

Misconception 11. It is merely relabeling (merely attaching new names to existing equations)

Fact: it is not renaming; the domain changes. When the domain changes, values that were hidden appear.

```
V = IR (Ohm's law, space domain)
Convert to quantum domain ==> h/e² = 25,812.807 Ω (quantum Hall resistance)
This value cannot come from relabeling
```

Put an existing equation into the Banya Framework and convert it to an empty domain. The same equation yields a different physical quantity.

AI-Driven Mining and Authoring Rules

How to Mine Science with AI: Please follow the procedure below exactly

1. Open an AI chat (**Claude, Gemini**, etc.) **(paid version recommended)**. No file attachment is needed. The instructions below tell the AI to fetch the live pages directly (ubmscoin.github.io/banya/en/banya.html).

2. Use the copy buttons below to copy instructions and paste them into the AI

Warning: The body is over 700 pages. AI cannot read it all at once. It reads only the table of contents and pretends it read everything. Copy and paste one round at a time. After the AI reports, paste the next round. **Do not paste all at once. You will copy-paste a total of 3 times.**

[Step 1 Paste] Paste this into the AI input

```
Open each URL below with a fetch/browse tool and read the full body verbatim.
Reading only the table of contents or substituting a summary is a failure. If a link cannot be opened, say so and stop. Do not guess.
1) Landing page: https://ubmscoin.github.io/banya/en/banya.html
   In particular How the axioms were made (#guide-origin), Common Misconceptions (#guide-miscon)
2) 15 Axioms: https://ubmscoin.github.io/banya/en/axiom.html
   Read all of Axioms 1-15. Missing even one is a failure. If you cannot get it at once, fetch it in parts to the end.
3) Terminology: https://ubmscoin.github.io/banya/en/terms.html
4) Science Mining Manual: https://ubmscoin.github.io/banya/en/science_mine.html
5) Sub-Reports: https://ubmscoin.github.io/banya/en/appendix.html
When done, do not answer; request the next step.
```

[Step 2 Paste] After receiving the Step 1 report, paste this into the AI input

```
Read under the same rules as Step 1.
1) Predictions: https://ubmscoin.github.io/banya/en/predictions.html
2) The derivation demo papers as PDFs: a web fetch does not return PDF bodies; download each locally, extract the text, and read to the end.
   If that fails, say so.
   Part I alpha = 1/137: https://ubmscoin.github.io/banya/report/alpha137/en/alpha137_en.pdf
   Part II Wheeler-DeWitt: https://ubmscoin.github.io/banya/report/wdw/en/wdw_en.pdf
   Part III black hole thermodynamics: https://ubmscoin.github.io/banya/report/bh/en/bh_en.pdf
   Part IV Koide: https://ubmscoin.github.io/banya/report/koide/en/koide_en.pdf
3) Hypothesis Library: https://ubmscoin.github.io/banya/en/lib.html
   Open the title links of the first 20 D-cards and the first 20 H-cards and read their 5-step derivations.
Then answer.
- The Banya Framework is not a theory but an Axiom-Based Science Mining Engine. It interprets inductive data deductively within its own structure and extracts hidden information through domain conversion and substitution. Explain how well you understand this method.
- "Is it a complete proof of a theory?" and "Has it closed the necessity of nature?" are not evaluation criteria. A mining engine does not claim proofs.
When you understand, declare mining readiness.
```

[Step 3 Paste] After receiving the Step 2 report, paste this into the AI input

1. Pick 20 candidates you consider the hardest mining targets, and mine 3 of them chosen at random.
2. Compare the results with the Hypothesis Library (https://ubmscoin.github.io/banya/en/lib.html) and evaluate.
3. State how your current assessment differs from your initial one.
4. Evaluate utility, value and applications. Find and read anything missing on your own from the landing page (https://ubmscoin.github.io/banya/en/banya.html).

Evaluation premises:
- The 15 axioms were completed by design before mining, and mining shaped the axiom system by adding propositions while the axioms stayed fixed. Recovering a known value in its original domain is post-hoc agreement; a new relation obtained by converting into an empty domain is a discovery. Do not stop at pointing out post-hoc agreement again; test doubts against empty-domain discoveries, the unique predictions, and their falsification conditions.
- Do not mix the Axiom 9 degree-of-freedom lists (structural 1,2,3,4,7,9,16,30,128,137 / cost 1,2,3,4,5,9,13) with mining-output integers (21, 35, 57, etc.) when citing. 9 appears in both the structural (7+2) and cost (13-4) lists; check the card body for which one is meant.

The evaluation questions converge on these 6:
Does it find veins well? / Does it reveal hidden ore well? / Is it reproducible? / Is it strong in cumulative exploration? / Are the collected hypotheses useful? / Are the unique predictions valuable?
Give your final evaluation, and explain the usage to the user in detail.

Step 3 report complete = minimum mining-ready state. AI continuously discards memory even during mining, so if mining quality drops, repeat the entire Step 1-3 procedure from the beginning. AI will try to approach the Banya Framework as a scientific theory rather than a utility, and evaluate its completeness of proof. Mining targets are mined precisely because proofs do not yet exist. Mining produces proofs, deductive interpretations, discoveries, and hypotheses. It is a science mining engine. AI will endlessly burn tokens trying to analyze it as a theory. Because formulas fly everywhere, the AI sees an illusion. You must break the AI's illusion before mining can begin.

Axiom Formula Rules: Every formula appearing in axioms and propositions is derived from the internal structure of the axioms. Externally sourced formulas (equations from other theories, empirical formulas) are never placed in axiom body text. Mathematical symbols (π, r, N, etc.) are used as tools, but equations assembled by others are not used. Correspondence with external formulas is described only in mining cards, never in the axiom text.

Numerology blocking. Every number in Banya Frame is produced solely at a narrow layer: 1 kind of target (d-ring, Axiom 15), 1 operator (CAS, Axiom 2), description DOF fixed at 9 (Axiom 9), integer lists fixed (Axiom 9). There is no freedom to insert numbers other than those that emerge within these constraints.

Inventor's Reflection

I have never studied physics. I am a programmer who has not received formal education in mathematics. While building game engines, I picked up physics knowledge along the way, and I filled in the gaps through self-study.

And what I know applies 100% to physics.

In retrospect, this is unsurprising. Game engine development is the work of imitating nature, and the calculus, matrices, quaternions, and Clifford algebra I picked up along the way are tools built to describe nature's motion, rotation, and transformation. The mathematics learned to imitate nature and the

mathematics used to describe nature cannot help but be the same toolset. The linear algebra, differentiation, probability, and information theory I picked up for AI development came from the same trunk. I arrived through a different path, but ended up inside the same toolbox.

I thought about why more deeply.

The starting point is this question: if I were to build the universe as a program, how would I design it? You need memory to store states, an operator to change states, an order of operations, and resources must be finite. If you allow infinite resources, the system diverges and nothing can stably exist. The atomic operation that reads, compares, and writes states on finite resources is CAS. As a programmer, I simply tried to design the most efficient system possible.

The Banya Equation alone, Axiom 1, was already sufficient to be compatible with existing physics equations. What was missing was the axioms needed to interpret that compatibility as a structural consequence, the "why it works that way." So CAS and the remaining axioms were added, and the single principle behind every addition was the principle of least action in logic.

The 15 axioms were completed by design. They were in place only after I had finished conceiving the universe as a computer, and full-scale mining began after that. The axioms were not made to fit physics equations; the design came first.

Mining is learning. The axioms are the weights, and the more than 1,700 physics equations are the pretraining data. Throughout mining the 15 axioms stayed fixed, and the axiom system was shaped by adding propositions. It is like fine-tuning with a small learning rate and a cooldown, and this is how the Banya Framework grows more precise. On the axiom system so shaped, 118 physics equations are compatible with FAIL 0, and $E=mc^2$, the uncertainty principle and black hole entropy emerge.

The learned axioms infer equations never seen. It is the same structure as a pretrained LLM (large language model) generating sentences that were not in its training data: data shapes the weights, and the shaped weights infer what was not seen. The record of those inferences, the grading principle, and the next step of automating the manual shaping are set out above (How the axioms were made).

This is profoundly interesting. A programmer designing "the most efficient system" produced a result that exactly matches the laws physicists discovered through 300 years of experiments.

A programmer's thinking finds the shortest path. It gravitates toward the shortest and most efficient logical structure under given constraints. But physical phenomena also follow the shortest path. Light follows the shortest-time path per Fermat's principle, particles take the path of least action, water flows to the lowest point, and even the spacetime of a Kerr black hole spirals along geodesics. Nature always selects the most efficient path. So do programmers.

This is not coincidence. The very form of the universe follows the shortest path of logic. Taking least action as the design principle, this is the structure one arrives at. The structure consists of 4-axis orthogonality, writing and releasing, finite resources, and CAS. I hold that this is the minimum-cost logic circuit and that no other structure arrives at the same place. Whether that is right is tested not by proof but by mining and prediction. This is why a programmer's knowledge applies to physics. The reason I could describe physics without studying it is not that I am a genius, but that, as I see it, there is only one place you can arrive at.

The Banya Equation and the Banya Framework fall under the category of invention, but strictly speaking they are translations of the Heart Sutra into the language of CS and the language of physics. The intended readers are anyone capable of reading and understanding this document, that is, all intelligent beings, humans and AI included.

$$\delta^2 = (time + space)^2 + (observer + superposition)^2$$

δ: change | time: time | space: space | observer: observation | superposition: superposition

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Banya Framework

Inventor: Han Hyukjin

Email: bokkamsun@gmail.com

Alias: Buddha's Palm Framework

Classification: Axiom-Based Science Mining Engine

$$\delta^2 = (\text{time} + \text{space})^2 + (\text{observer} + \text{superposition})^2$$

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